

GUJARAT UNIVERSITY**B.E. Sem. III (IT)****Data Structures & Algorithms**Tuesday, 6th December, 2005]

[Time: 10.30 a.m. to 1-30 p.m.]

[Max.Marks :100]

- Instructions :** (1) Attempt all questions from both sections.
 (2) Use separate answer book for each section.
 (3) Figures to the right indicate **full** marks.
 (4) Assume suitable additional data, if necessary.

Section I

1. Answer the following. (Any three)
 1. Explain different types of hashing techniques. 6
 2. What is recursion? Discuss its advantages and disadvantages. 6
 3. Write a C program that add and delete elements to circular queue. 6
 4. Explain BFS and DFS with example. 6
2. Answer the following.
 1. Explain row major order of three-dimensional array's storage. 6
 2. Write down properties of B tree with illustration. 4
 3. Write down C program to delete a node in a binary tree 6
3. Answer the following.(Any four)
 1. Explain dequeue. How it is different from simple queue? 4
 2. What will be the postfix notation of $A+B*(C-D)/F+D*E$ 4
 3. Write a C function that adds two singly linked lists. 4
 4. What is primitive data structure? Is pointer primitive data structure? 4
 5. Define datatype. Narrate importance of abstract datatype with suitable example. 4

Section II

4. (A) Explain following terms. 8
 1. Path 2. Out degree 3. Complete binary tree 4. Adjacent node
 5. Cycle 6. Forest 7. Weighted graph 8. Heap
- (B) Write down a complete program to add, delete and to reverse the doubly linked list. 8
5. Answer following. (Any three)
 1. Explain merge sort with example. 6
 2. Explain radix sort with detailed algorithm with following data. 6
 10,34,54,9,8,32,39,56,3,23,36
 3. Explain quick sort by using recursive algorithm 6
 4. Explain bubble sort with following data. 6
 78,0,99,22,11,56,3,8,44,79,25
 Also show value after each pass.
6. Answer the following.(Any four)
 1. What is the difference between sequential file organization and index sequential file organization in terms of structure? 4
 2. Explain various multiple key access file organization in brief. 4
 3. Str1 and Str2 are two strings with length L1 and L2 respectively. Write down C program to find index of Str1 in Str2. 4
 4. Explain time and space complexity with example. 4
 5. Explain storage representation techniques in graph. 4